

EDUCATION

Master of Science in Mechanical Engineering – Penn State
Bachelor of Science in Mechanical Engineering – Penn State
Minor in Entrepreneurship and Innovation – Penn State

2025–present, GPA: 4.0/4.0
2021–2025, GPA: 3.92/4.0
2021–2025, GPA: 4.0/4.0

TECHNICAL EXPERIENCE

Robotics & Automation Systems

- Designed a UR5e robotic assembly system integrating RTDE robot control, LLM-based reasoning, and VLM-based perception.
- Developed an RRT-based Model Predictive Control framework for path planning and trajectory optimization

AI-Enabled Robotic Agents

- Designed an LLM-enabled robotic agent that interprets and converts human instructions into robot actions.
- Fine-tuned and machine-learning models to improve context understanding, task reasoning, and robotic agent behavior.
- Developed multi-agent planning logic for adaptive task allocation, coordination, and execution across robotic systems.

Computer Vision

- Developed an adaptive vision system combining 3D point-cloud processing with VLM reasoning for novel-object identification.
- Designed and evaluated object detection models, including YOLO and SAM-based methods.

PUBLICATION

COVILLM: An Adaptive Human-robot Collaborative Assembly Framework Using Large Language Models

Speech-to-Trajectory: An LLM-enabled Human-Robot Collaborative Framework for Generating New Assembly Tasks from Verbal Guidance

Adaptive Task Planning and Coordination in Multi-Agent Manufacturing Systems Using Large Language Models.

INTERSHIP

Internship – PennDOT Automated Driving System

- Developed and test software systems using ROS to support message passing between network sensors.
- Created MATLAB simulations to model and control autonomous vehicle systems.
- Designed, built, and validated integrated camera, LIDAR, and radar systems for autonomous vehicle applications.

Hydraulic Submergence Project – Flowserve US Inc

- Designed and developed a hydraulic test rig to investigate vortex formation and behavior in an upside-down pump configuration.
- Conducted CFD/FEA simulations and DFMEA to optimize design parameters and ensure system reliability.

Entrepreneurship and Innovation

- Learned the foundation of starting a professional business through income statements, balance sheets and cash flow.
- Delivered an investment pitch about product branding, marketing strategies, financial forecasting, and accounting principles.
- Founded a startup, Just-EnCase, introducing an innovative container with suction cup technology for personal storage solutions.

LEADERSHIP

Vice President — Asian Student Fitness Club (ASFC)

- Dedicating to creating a vibrant and supportive community for students passionate about fitness and bodybuilding.
- Collaborated with other clubs and local businesses to organize engaging events and opportunities for members.

Program Leader — Summer Bridge Engineering Program

- Directed a summer program in collaboration to educate and facilitate a sense of community among 60 engineering students
- Collaborated with fellow program assistants to strategize program plans that creating optimal and equal learning experience

AWARD

Mechanical Engineering Graduate School Scholarship

- Scholarship for individuals for top percentile of a very talented class of students

Cohort 2 Scholar — The Pennsylvania State University A. James Clark Scholars Program

- The scholarship covers the cost of tuition, fees, and room and board, as well as cocurricular enrichment programs

Harding Louis Mem Scholarship

R&M Munro Trustee Scholarship

SKILLS

Python

Machine-learning

Fine-tuning

ROS

LLM

VLM

MATLAB

EXCEL

Simulink

Solidwork

OpenCAD

Minitab

PowerPoint

Gurobi

C++